

Identification of Asali (Genuine) or Nakali (Counterfeit) Items

Project Report Part A

Information Integration and Application (CSE656)

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Abstract

This project implements a federated data-integration system that identifies whether pharmaceutical products are **Asali** (genuine) or **Nakali** (counterfeit). The system reconciles data across four isolated entities: Manufacturer, Distributor, Vendor, and the Ministry. By discovering shared schema connections without prior shared keys, executing parallel API calls, and evaluating cross-source integrity rules, the system generates an auditable authenticity verdict and provenance timeline.

1 Scope of Work

1.1 In Scope

- **Four Independent Databases:** Designed completely in isolation with distinct naming styles and zero shared foreign keys.
- **REST APIs & Mediator Architecture:** Every database is fronted by an independent API; a mediator handles queries exclusively through these endpoints without direct database access.
- **Instance-Based Schema Matching:** Discovers matching attributes across isolated schemas based on data-value overlap.
- **Multi-Machine Deployment:** Works across one to four machines via configuration without code changes.
- **Rule-Based Verification Engine:** Evaluates cross-source consistency to produce verdicts (ASALI, SUSPECT, NAKALI), trust scores, and provenance histories.
- **Automated Write-Back Enforcement:** Counterfeit reports submitted to the Ministry trigger an automated API call to blacklist the offending supplier at the Distributor.
- **GUI Interface:** A clean web dashboard that displays results, value conflicts, and query execution plans.

1.2 Out of Scope

Physical barcode hardware, real GS1/blockchain infrastructure, production auth/RBAC, and direct integration with live government portals.

2 New and Innovative Aspects

1. **Automated Join Key Discovery:** The system uses Jaccard set overlap on actual values to automatically identify that differing column names refer to the same item code.
2. **Federated Anomaly Detection:** Uncovers fakes by detecting inconsistencies across databases, such as cloned serials (quantity over-issue), broken custody chains, and ghost batches.
3. **Auditable Trust Scoring:** Decisions are transparent; each triggered rule applies an explicit numerical penalty or reward.

4. **Closed-Loop Enforcement:** Directly connects regulatory reporting to distributor supplier blacklisting through source-to-source API communication.
5. **Transparent Query Execution:** The GUI exposes exact sub-queries, execution times, and target sources for full inspection.
6. **Location-Transparent Mesh:** The mediator uses dynamic service discovery, running identically whether local or distributed across multiple physical machines.

3 Database Schema Design

Four isolated SQLite databases were created with distinct naming conventions and structures to replicate real-world data silos.

1. Manufacturer Database (manufacturer.db) — snake_case

- **companies** **PK** company_id | company_name | license_no | country | established_year
- **products** **PK** product_id | **FK** company_id | brand_name | generic_name | dosage_form | strength
- **manufactured_batches** **PK** batch_id | **FK** product_id | **UNIQUE** gtin_serial | mfg_date | expiry_date | batch_qty | mrp | factory_location

2. Distributor Database (distributor.db) — Abbreviated prefixes

- **suppliers** **PK** sup_id | sup_name | drug_license | gtin | region | contact_email | blacklist_flag
- **inbound_consignments** **PK** consignment_id | **FK** sup_id | source_company | received_on | invoice_no
- **distributed_stock** **PK** dist_row_id | **FK** consignment_id | item_code | product_desc | qty_supplied | unit_price | supplied_to_vendor | dispatch_date

3. Vendor Database (vendor.db) — camelCase

- **vendors** **PK** vendorId | vendorName | shopLicense | city | pincode
- **purchaseScans** **PK** scanId | **FK** vendorId | productBarcode | scannedLabelName | packSize | sellingPrice | scanTimestamp
- **customerSales** **PK** saleId | **FK** scanId | saleTime | billNo | customerPhoneHash

4. Ministry Database (ministry.db) — Business reference keys

- **counterfeit_reports** **PK** report_ref | suspect_code | reported_by_vendor | reported_against_supplier | complaint_date | lab_result | status
- **verified_genuine_registry** **PK** reg_id | **UNIQUE** auth_code | verifying_authority | verified_on

- **enforcement_actions** **PK** action_id | **FK** report_ref | action_taken | penalty_amount | action_date

3.1 Cross-Source Attribute Alignments

The table below illustrates the semantic mappings identified across the isolated schemas.

Concept	Manufacturer	Distributor	Vendor	Ministry
Item unique code	gtin_serial	item_code	productBarcode	suspect_code / auth_code
Company name	company_name	source_company	—	reported_against_supplier
Supplier name	—	sup_name	—	reported_against_supplier
Vendor name	—	supplied_to_vendor	vendorName	reported_by_vendor
Price	mrp	unit_price	sellingPrice	—

4 Populating the Data

The databases are populated with comprehensive test records: 3 companies, 10 products, 10 batches, 3 suppliers, 5 consignments, 11 distribution rows, 3 vendors, 13 scans, over 140 sales, and regulatory records. Twelve explicit validation scenarios test the rule engine:

Sample Code	Verdict	Integrity Rule Evaluated
SLP-AMOX500-B2401-000	ASALI	Verified genuine; consistent end-to-end supply chain
SLP-PANT040-B2402-002	NAKALI	Confirmed counterfeit by Ministry laboratory testing
MRC-DOLO650-B2403-004	NAKALI	Chain of custody broken (dispatched to CityCare, sold at QuickMeds)
SLP-VITD3-B2404-0050	NAKALI	Cloned serial code: batch size 50, but 105 units scanned and sold
APX-COUGH100-B2311-00	NAKALI	Expired stock: unit sold past labeled expiry date
SLP-INSUL-B2406-0033	SUSPECT	Origin mismatch: distributor source company differs from manufacturer
ZZZ-CIPRO500-FAKE-900	NAKALI	Ghost batch: serial number absent from manufacturer records

5 APIs Creation and Schema Matching

5.1 REST API Microservices

Each database is wrapped in a dedicated FastAPI service exposing standardized endpoints:

- GET /schema and GET /sample: Return metadata and distinct sample values for schema discovery.
- POST /query: Executes parameterised, injection-safe read queries returning data and

executed SQL.

- `POST /reports` & `POST /flag-supplier`: Support write-back operations for dispute logging and supplier blacklisting.

5.2 Instance-Based Schema Matching

Column equivalence is calculated using a weighted similarity metric combining structure and content:

$$\text{score} = 0.30 \cdot \text{name_similarity} + 0.10 \cdot \text{type_similarity} + 0.60 \cdot \text{value_similarity}$$

Value similarity evaluates the Jaccard overlap of sample values retrieved via `GET /sample`. This heavy weighting allows the matcher to identify that `gtin_serial`, `item_code`, `productBarcode`, and `suspect_code` represent the same global join key.

6 Query Decomposition and Federation

The mediator maintains zero local storage. When querying an item, it performs parallel decomposed fetches using a thread pool:

- **Round 1:** Dispatches filtered queries for the given item code concurrently to all four services.
- **Round 2:** Uses primary/foreign keys returned from Round 1 to retrieve batch, consignment, vendor, and report records.
- **Round 3:** Fetches high-level parent company and supplier profile details.
- **Synthesis:** Joins all records in Python to form a complete provenance history, evaluate integrity rules, and generate the final verdict.

7 Communication Between Data Sources

All inter-service traffic uses JSON over HTTP.

- **Network Topology:** Supports running all services on a single host or distributed across two to four physical machines without modifying code.
- **Service Discovery:** Handled via `sources.json` and the `service_url()` resolver, falling back cleanly to local defaults when co-located.
- **Direct Source-to-Source Enforcement:** When a counterfeit report is logged at the Ministry service, it autonomously issues an HTTP call directly to the Distributor API to set `blacklist_flag = 1` on the supplier.
- **Fault Handling:** Network failures return clear HTTP 502 error structures rather than unhandled server crashes.

8 Query Results Integration and GUI

The mediator hosts a web UI delivering complete transparency into the integration pipeline:

- **Complete History View:** Displays an immediate status banner (ASALI, SUSPECT, or NAKALI), numerical trust score, active red flags, and an end-to-end timeline color-coded by originating data source.
- **Cross-Source Discrepancy Tracking:** Delineates between normal price variations (whole-sale vs. retail) and semantic conflicts (mismatched supplier/manufacturer).
- **Interactive Query Planner:** Displays the exact decomposition plan, including target host, SQL statement, and network latency for each sub-query.
- **Integration Lab:** Visualizes the schema matching similarity matrix, discovered column clusters, and GAV view mappings.
- **Direct Enforcement Trigger:** Enables users to file counterfeit reports with one click, updating the regulatory registry and blacklisting suppliers in real time.